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Roll No. ....

Total No. of Questions : 06] [Total No. of Printed Pages : 04

ZE-193

B. Tech. (New Scheme) (BE) (Main/ATKT)

Examination, 2026

(Fourth Semester)

ANALOG CIRCUITS

EL-402

Time : 3 Hours] **[Maximum Marks : 60**

Note : Attempt all questions. All questions carry equal marks.

1. Discuss the concept of the operating point (Q-point) in transistor biasing.

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Note : Attempt all questions. All questions carry equal marks.

1. Discuss the concept of the operating point (Q-point) in transistor biasing.

Or

Derive the expressions for collector current and collector-emitter voltage in a voltage-divider biased BJT.

2. What is a multi-stage amplifier ? Explain its need and working with the help of a block diagram.

Or

Explain the construction and operation of a Class A power amplifier with a neat circuit diagram. Mention its advantages and disadvantages.

3. What is an oscillator ? Explain its working principle and the conditions required for sustained oscillations.

Or

Describe the working of a Wien Bridge Oscillator. Mention its advantages, disadvantages, and applications.

4. What is an Operational Amplifier (Op-Amp) ? Explain its construction, characteristics and working with a block diagram.

Or

Explain the ideal characteristics of an operational amplifier. How do they differ from the practical characteristics ?

5. Explain the block diagram of an ADC. Describe the function of each block.

Or

Describe the working of a Flash ADC. Discuss its advantages and disadvantages

6. Explain the concept of negative feedback in operational amplifiers. What are its advantages ?

Or

Write short notes on the following :

(a) CMRR

(b) Slew rate.

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